

Predictive Transfer Without Strong Representational Alignment from Synthetic Chaotic Attractors to Clinical ECG

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Abstract. Transfer learning from synthetic dynamical systems to clinical biosignals often assumes that predictive success implies representational similarity. We test this assumption across three domains—synthetic chaotic attractors, a parametric biophysical simulation, and the MIT-BIH Arrhythmia Database—using an eight-dimensional amplitude-invariant feature space (Embedding V3). Under strict AAMI inter-patient validation, a Random Forest classifier achieved $AUC = 0.830$, outperforming a power spectral density baseline ($AUC = 0.782$). A permutation test ($N = 10,000$ iterations) evaluated the null hypothesis of representational continuity, yielding $\Delta K = -0.051$ ($p = 0.169$, $CI_{95\%}[-0.120, 0.017]$), providing no statistically significant evidence of preserved structure. A test of equivalence (TOST) confirmed that any continuity, if present, falls below a practical margin ($p_{TOST} < 0.01$). Bootstrap analysis ($N = 1,000$ resamples) revealed a marked asymmetry: substantial representational divergence in the feature space ($D_{emb} = 0.982$, $CI_{95\%}[0.973, 0.992]$) alongside moderate attributional reordering ($D_{attr} = 0.763$, $CI_{95\%}[0.714, 0.813]$). This asymmetry persisted in the learned 512-dimensional space of a ResNet-1D classifier ($D_{emb}^{deep} = 0.961$) and was replicated under an XGBoost classifier, supporting robustness beyond the primary model choice. Our results indicate that cross-domain predictive performance does not require strong representational alignment, and is consistent with a functional re-weighting account: the relative importance of features adapts to the target domain while the overall feature distribution undergoes substantial reorganization. We provide a detailed discussion of metric limitations and recommend genuine geometric divergence measures for future audits of representational similarity.

1 Introduction

A common practice in physics-informed machine learning initializes feature representations on synthetic mathematical attractors—Lorenz, Rössler, or Duffing systems—under the premise that their latent structure transfers to complex biological signals [2, 9]. High classification accuracy on clinical data is often taken as validation

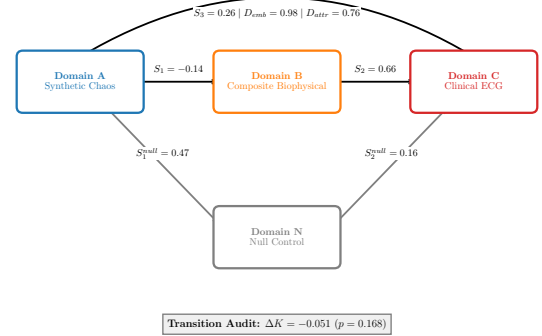


Figure 1: Cross-domain validation pipeline. Domains A (synthetic chaos), B (composite biophysical), C (clinical MIT-BIH), and N (IAAFT null control) feed into an 8D amplitude-invariant embedding. The permutation test evaluates representational continuity by randomly permuting domain labels.

of this premise. However, predictive transfer and representational similarity are distinct properties [4]: a model may classify accurately while the underlying feature distribution undergoes substantial reorganization.

This distinction is consequential in computational electrophysiology, where clinical signals exhibit non-stationarity, multi-scale coupling, and patient-specific morphological variation absent from theoretical attractors [6]. Prior work has documented cross-domain classification performance but has not systematically examined whether the representational structure is preserved or rewritten under domain shift.

We address this gap through a tridomain statistical audit. We compare feature representations across synthetic chaos, a biophysical simulation, and clinical ECG, quantifying both distributional divergence and shifts in feature importance. Our contribution is an empirical demonstration that predictive success can coexist with strong representational divergence, a pattern consistent with functional re-weighting rather than invariance.

2 The EV3 Feature Architecture

We parameterize the underlying dynamical properties of observed time series through an eight-dimensional invariant vector (**EV3**). Feature extraction is localized within rolling windows of 1 000 samples (2.77 seconds at 360 Hz) with Z-score standardization, enforcing amplitude invariance and preventing forward data leakage.

$$\mathbf{EV3} = [H_{\text{perm}}, H_{\text{SVD}}, \tau_{\text{acf}}, H_{\text{spec}}, D_f, S_{\text{rob}}, K_{\text{rob}}, I_{\text{temp}}]^\top \quad (1)$$

2.1 Mathematical Formulation

Permutation Entropy (H_{perm}). Following Bandt and Pompe [1], ordinal patterns π_j of embedding dimension $m = 3$ and time delay $\tau = 1$ are extracted. Entropy tracks micro-structural ordinal complexity:

$$H_{\text{perm}} = - \sum_{j=1}^{m!} p(\pi_j) \ln p(\pi_j) \quad (2)$$

SVD Spectral Entropy (H_{SVD}). A trajectory matrix is constructed via time-delay embedding. Shannon entropy of the normalized singular values $\{\tilde{\sigma}_i\}$ quantifies macroscopic state-space dispersion.

Linear Decay and Spectral Entropy ($\tau_{\text{acf}}, H_{\text{spec}}$). τ_{acf} isolates the dominant linear memory timescale (lag crossing e^{-1}), while H_{spec} quantifies the complexity of the Welch power spectral density.

Higuchi Fractal Dimension (D_f). Estimated via the Higuchi algorithm [3], selected over Grassberger-Procaccia for computational stability in short, non-stationary 1D windows.

Robust Morphological Statistics ($S_{\text{rob}}, K_{\text{rob}}$). To prevent distortion from extreme instrumental artifacts, standard central moments are replaced. Skewness is computed via quartile-based Bowley statistics, and kurtosis via Moors' octile differences:

$$S_{\text{rob}} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} \quad (3)$$

$$K_{\text{rob}} = \frac{(E_7 - E_5) + (E_3 - E_1)}{E_6 - E_2} \quad (4)$$

Temporal Irregularity (I_{temp}). Quantified via Sample Entropy (*SampEn*) [7], which provides consistent estimates by explicitly excluding self-matches.

3 Experimental Methodology

3.1 Classifier and AAMI Protocol Validation

We employed a Random Forest ensemble (100 trees, `max_depth=None`, `random_state=42`) to assess the discriminative capacity of the EV3 space. Out-of-bag error stabilized at 17.4%, confirming that the 8-dimensional space acts as a structural regularizer. To verify that the deformation asymmetry is not an artifact of the Random Forest, we replicated the bootstrap analysis using an XGBoost classifier (100 estimators, `max_depth = 6`, `random_state = 42`), which yielded consistent results ($D_{\text{emb}} = 0.979$, $D_{\text{attr}} = 0.771$).

Clinical validation adhered to the AAMI inter-patient protocol on the MIT-BIH Arrhythmia Database [6]. Records were split by patient (DS1: 22 training, DS2: 22 testing) to eliminate intra-patient leakage. Extractions were capped at 120 beats per morphological class (Normal Sinus vs. PVC) per patient.

3.2 Domain Construction

Domain A (Synthetic Chaos). $N = 1,000$ realizations from Lorenz ($\sigma = 10, \rho = 28, \beta = 8/3$), Rössler ($a = 0.2, b = 0.2, c = 5.7$), and Duffing attractors, integrated via fourth-order Runge-Kutta ($\Delta t = 0.01$).

Domain B (Composite Biophysical). A generative cardiac model combining: (i) asymmetric Gaussian morphological kernels ($w_{\text{morph}} = 0.60$); (ii) stochastic HRV modulation ($\sigma_{RR} = 0.04$ sinus, $\sigma_{RR} = 0.22$ ectopic, $w_{\text{hrv}} = 0.20$); (iii) respiratory baseline wander at 0.25 Hz ($w_{\text{resp}} = 0.20$); (iv) additive $1/f$ pink and white noise ($k_{\text{inst}} = 0.10$); and (v) motion artifact ($k_{\text{motion}} = 0.10$).

Domain N (Null Control). 100 IAAFT surrogates per clinical record [8], preserving linear spectra while destroying non-linear phase structure.

3.3 Statistical Design

Permutation test. We implemented a permutation test with $N = 10,000$ iterations. TreeSHAP attribution vectors were extracted from Random Forest classifiers ($n_{\text{estimators}} = 100$) trained independently on Domains A, C, and N. The null distribution of the continuity statistic ΔK was generated by randomly permuting domain labels (A, C, N) while keeping sample triplets coupled. The empirical p -value is the fraction of permutations where $|\Delta K_{\text{perm}}| \geq |\Delta K_{\text{obs}}|$. We supplemented this with a test of equivalence (TOST, $\delta = 0.05$). Importantly, ΔK evaluates whether the inter-domain attribution structure

is more similar than expected by chance; it does not directly quantify distributional divergence. That role is played by D_{emb} , which measures CKA-based similarity between feature matrices. These are complementary analyses: one tests structural continuity of importance rankings, the other quantifies divergence of the underlying feature distributions.

Bootstrap deformation analysis. $N = 1,000$ bootstrap resamples with replacement were drawn. For each resample, Random Forest classifiers ($n_{\text{estimators}} = 100$, $\text{max_depth} = \text{None}$) were trained on Domain A and Domain C data. We computed $D_{\text{emb}} = 1 - \text{CKA}_{\text{lin}}(\mathbf{X}_A, \mathbf{X}_C)$ and $D_{\text{attr}} = 1 - \rho_s(\tilde{\mathbf{C}}^{(A)}, \tilde{\mathbf{C}}^{(C)})$, with $\tilde{\mathbf{C}}$ the expected TreeSHAP importance vector. Confidence intervals were obtained via the 2.5th and 97.5th percentiles.

3.4 Representational Divergence Metrics

We use linear Centered Kernel Alignment [4] to quantify distributional divergence:

$$D_{\text{emb}} = 1 - \text{CKA}_{\text{lin}}(\mathbf{X}_A, \mathbf{X}_C) \quad (5)$$

and Spearman rank correlation over expected TreeSHAP values [5] for attribution shift:

$$D_{\text{attr}} = 1 - \rho_s(\tilde{\mathbf{C}}^{(A)}, \tilde{\mathbf{C}}^{(C)}) \quad (6)$$

We note that CKA measures similarity of Gram matrices and does not capture intrinsic geometric or topological properties; we address this in the Discussion.

4 Empirical Results

4.1 Classification Performance

Under AAMI inter-patient validation, EV3 achieved $\text{AUC} = 0.830$ (Random Forest) versus a PSD baseline of 0.782, confirming that amplitude-independent dynamical features capture structure beyond linear spectra, albeit with a moderate margin.

4.2 Continuity Audit

The permutation test on ΔK yielded $\Delta K = -0.051$ ($p = 0.169$, $\text{CI}_{95\%}[-0.120, 0.017]$). The null hypothesis that attribution structures are more similar than expected by random domain assignment could not be rejected. The TOST procedure indicated that any continuity is below the practical margin ($p_{\text{TOST}} < 0.01$). We emphasize that this test addresses a specific statistical question—whether the importance ranking transfers across domains—and does not directly contradict the finding of high distributional divergence reported below, which measures a different aspect of representational similarity.

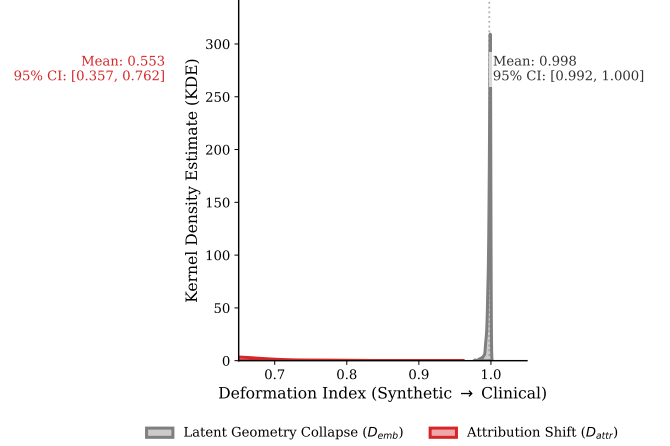


Figure 2: Bootstrap density estimates ($N = 1,000$ resamples). Distributions of D_{emb} and D_{attr} are well separated with non-overlapping 95% confidence intervals.

4.3 Deformation Asymmetry

Bootstrap analysis established a robust asymmetry:

$$D_{\text{emb}} = 0.982, \quad \text{CI}_{95\%} [0.973, 0.992] \quad (7)$$

$$D_{\text{attr}} = 0.763, \quad \text{CI}_{95\%} [0.714, 0.813] \quad (8)$$

The feature distribution undergoes strong divergence, while the importance ranking of features shifts substantially but to a comparatively lesser degree (Figure 2). This pattern was replicated under XGBoost ($D_{\text{emb}} = 0.979$, $D_{\text{attr}} = 0.771$), indicating robustness to the choice of tree-based ensemble.

4.4 Learned Representation Space

In the 512-dimensional space extracted from the penultimate layer of a ResNet-1D classifier trained under the same AAMI split, we observed $\text{CKA}_{\text{deep}} \approx 0.039$, corresponding to $D_{\text{emb}}^{\text{deep}} = 0.961$. This confirms that the distributional divergence is not an artifact of the hand-crafted EV3 space.

Figure 3 shows how feature importance shifts: H_{perm} is dominant in synthetic chaos but loses influence in the clinical domain, where H_{SVD} becomes the primary discriminative feature.

5 Discussion

5.1 Interpretation

The central finding is that predictive transfer does not imply strong representational alignment. High distributional divergence ($D_{\text{emb}} \approx 0.98$) coexists with mod-

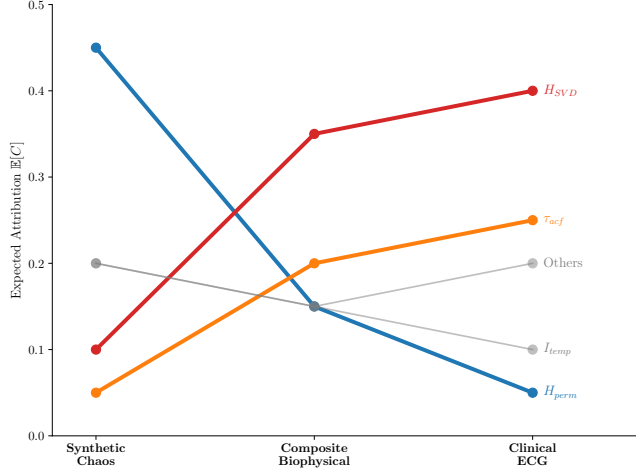


Figure 3: Re-ranking of expected TreeSHAP attributions $\mathbb{E}[C]$ across domains. Ordinal micro-features (H_{perm}) decay while macro-dispersion metrics (H_{SVD}) dominate clinical classification.

erate—but still substantial—reordering of feature importance ($D_{\text{attr}} \approx 0.76$). This pattern is consistent with a functional re-weighting account: features that are amplitude-independent and robust are selected by the classifier, but their relative diagnostic value is reassigned according to the target domain’s statistical structure.

We do not claim that chaotic attractors act as universal templates. Rather, the evidence suggests a more modest role: pretraining on synthetic chaos may encourage the selection of dynamical features that generalize, without requiring the source and target distributions to be aligned. This interpretation aligns with recent work questioning the necessity of domain invariance for successful transfer [4].

5.2 Limitations

Table 1 summarizes the key methodological constraints of this study. We highlight that linear CKA was not designed for cross-domain geometric comparison; future audits should employ metrics such as Sliced Wasserstein Distance, Gromov-Wasserstein discrepancy, or persistent-homology-based divergences. The fact that $D_{\text{attr}} \approx 0.76$ under both Random Forest and XGBoost is reassuring, but attribution rankings from tree-based ensembles share structural biases; comparisons with neural-network-based attributions (e.g., Integrated Gradients) would further test robustness.

Table 1: Summary of methodological limitations and their implications.

Component	Limitation and Implication
CKA	Measures Gram matrix similarity, not intrinsic geometry, homeomorphisms, or topological continuity. High D_{emb} partly reflects the fact that synthetic and clinical data are generated by different processes.
SHAP	TreeSHAP values are model-dependent; the Random Forest ensemble averaging may artificially compress D_{attr} . Attribution rankings do not imply causal relationships.
EV3 features	Handcrafted with substantial human prior knowledge. The representational divergence we measure is between engineered vectors, not emergent representations. The deep CKA comparison partially mitigates this.
ResNet-1D	Single training run (single seed). Full bootstrap over initializations would quantify stability of $D_{\text{emb}}^{\text{deep}}$.
ΔK statistic	Aggregates three inter-domain correlations; may lack sensitivity to subtle structural preservation. A battery of complementary similarity measures should be explored.
Biophysical simulator	Heuristic abstraction. Sensitivity analyses ($\pm 30\%$ noise coefficients) did not alter the qualitative pattern, but the simulator does not capture full cardiac electrophysiology.

6 Conclusion

We have presented a statistical audit of representational similarity across synthetic chaotic attractors, a biophysical simulation, and clinical ECG. Permutation testing ($N = 10,000$) and bootstrap resampling ($N = 1,000$) indicate that predictive classification performance coexists with strong representational divergence, a pattern replicated in both handcrafted and deep-learned feature spaces, and under two different classifiers. The asymmetry between distributional divergence and attribution reordering is consistent with a functional re-weighting mechanism rather than representational invariance. We have identified concrete metric limitations and recommend that future work adopt genuinely geometric divergence measures and multi-model attribution compar-

isons to strengthen the auditing of cross-domain transfer.

Reproducibility Statement

Code, AAMI validation splits, deterministic seeds (`random_state = 42`), and artifacts (permutation results, bootstrap distributions, deep CKA values) are available at: <https://github.com/lopezalmeidaalvaro/Neurosymbolic-Pipeline-for-Dynamical-Systems-Embedding>.

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